

Vartika Singh, PhD

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Research professional in agricultural and environmental economics with expertise in assessment of sustainable food systems, agricultural policy, and climate impact research in South Asia. I specialise in integrated assessment modeling and policy analysis for sustainable transformation. I play a critical role in generating evidence for policy formulation and translating complex scientific research into actionable policy insights for decision-makers. I have a proven track record in leading international research collaborations, coordinating with stakeholders across various global conventions (including UNFCCC, CBD, and SDGs), and publishing in high-impact journals. I am experienced in bridging the science-policy interface through strategic engagement with government agencies, international organizations, and research institutions.

Employment History

June 2026 –	📌 Associate Research Fellow International Food Policy Research Institute, India.
June 2024 – May 2026	📌 Sr Research Analyst II International Food Policy Research Institute, India.
2019 – 2024	📌 Sr Research Officer Indian Institute of Management, Ahmedabad.
2017 – 2019	📌 Program Manager International Food Policy Research Institute, India.
2015 – 2017	📌 Research and Training Manager EPoD, Harvard Kennedy School, India
2012 – 2015	📌 Research Analyst International Food Policy Research Institute, India.
2011 – 2012	📌 Program Associate ICICI Foundation for Inclusive Research.

Education

2019 – 2024	📌 PhD Agricultural Economics , Humboldt University of Berlin. Defense: June 2025 Thesis title: <i>Pathways to Sustainable Transformation in India by 2050: A Food-Water-Land Nexus Approach</i>
2009 – 2011	📌 Master's in Development Studies Tata Institute of Social Sciences, Mumbai, India.
2006 – 2009	📌 Bachelor's in Economics Hindu College, Delhi University, India.

Detailed Publication List

High Impact - Q1 as per Scimago Journal Rankings

- 1 Bodirsky, Benjamin, Felicitas Beier, Florian Humpenöder, **Vartika Singh**, et al. "A food system transformation pathway reconciles 1.5°C global warming with improved health, environment and social inclusion." *Nature Food*, 2025.
- 2 **Singh, Vartika**, Miodrag Stevanovic, et al. "A comparison of the effects of local and EAT-Lancet dietary recommendations on selected economic and environmental outcomes in India." *Food Policy*, 2025.
- 3 zIris, Berger, Manan Bhan, Rekha Bhangaonkar, Abhishek Chaudhary, **Vartika Singh**, et al. "Integrated land systems for sustainable food production and biodiversity conservation in the semi-arid to moist tropics: stakeholder perspectives from Andhra Pradesh, India." *Frontiers in Sustainable Food Systems*, 2025.
- 4 Perez, Nicostrato, **Vartika Singh**, Claudia Ringler, Hua Xie, Tingju Zhu, Edwin H Sutanudjaja, and Karen Vilholth. "Ending groundwater overdraft without affecting food security." *Nature Sustainability*, 2024.
- 5 Bustamante, Mercedes, Joyashree Roy, Daniel Ospina, **Vartika Singh**, et al. "Ten new insights in climate science 2023/2024." *Global Sustainability*, 2023.

- 6 Mosnier, Aline, Valeria Javalera-Rincon, Sarah Kate Jones, Robbie Andrew, Zhaohai Bai, Justin Baker, **Vartika Singh**, et al. "A decentralized approach to model national and global food and land use systems." *Environmental Research Letters*, 2023.
- 7 **Singh, Vartika**, Miodrag Stevanovic, Chandan Kumar Jha, Felicitas Beier, Ranjan Kumar Ghosh, Hermann Lotze-Campen, and Alexander Popp. "Assessing policy options for sustainable water use in India's cereal production system." *Environmental Research Letters*, 2023.
- 8 Urfels, Anton, Kai Mausch, **Vartika Singh**, et al. "Farm size limits agriculture's poverty reduction potential in Eastern India even with irrigation-led intensification." *Agricultural Systems*, 2023.
- 9 Jha, Chandan Kumar, Ranjan Kumar Ghosh, **Vartika Singh**, et al. "Pathway to achieve a sustainable food and land-use transition in India." *Sustainability Science* 18, no. 1 (2022).
- 10 Jha, Chandan Kumar, **Vartika Singh**, et al. "The role of food and land use systems in achieving India's sustainability targets." *Environmental Research Letters* 17, no. 7 (2022).
- 11 Mosnier, Aline, Guido Schmidt-Traub, Obersteiner, **Vartika Singh**, et al. "How can diverse national food and land-use priorities be reconciled with global sustainability targets? Lessons from the FABLE initiative." *Sustainability Science* 18, no. 1 (2022).
- 12 Ghosh, Ranjan Kumar, Shweta Gupta, **Vartika Singh**, and Patrick S Ward. "Demand for crop insurance in developing countries: new evidence from India." *Journal of Agricultural Economics* 72, no. 1 (2021).
- 13 Gupta, Shweta, Avinash Kishore, Muzna Fatima Alvi, and **Vartika Singh**. "Designing better input support programs: Lessons from zinc subsidies in Andhra Pradesh, India." *PLoS One* 15, no. 12 (2020).
- 14 Ward, Patrick S, Shweta Gupta, **Vartika Singh**, et al. "What is the intrinsic value of fertilizer? Experimental value elicitation and decomposition in the hill and terai regions of Nepal." *Food Policy* 90 (2020).
- 15 Ward, Patrick S, and **Vartika Singh**. "Using field experiments to elicit risk and ambiguity preferences: Behavioural factors and the adoption of new agricultural technologies in rural India." *The Journal of Development Studies* 51, no. 6 (2015).
- 16 Ward, Patrick S, David L Ortega, David J Spielman, and **Vartika Singh**. "Heterogeneous demand for drought-tolerant rice: Evidence from Bihar, India." *World Development* 64 (2014).

In Review/Pipeline

- 1 Gupta, Shweta, Prapti Barooah, **Vartika Singh**, Muzna Alvi, Ruth Meinzen-Dick, and Claudia Ringler. "Hello! Can you hear me? Impact of speakerphones on phone survey responses." *Under Preparation for submission to Political Science Research and Methods*, 2026.
- 2 Jessica Fanzo, Benjamin Bodirsky, Marco Springmann, and **Vartika Singh**. "Ten new insights in climate science 2026." *Under Preparation for submission to Global Sustainability*, 2026.
- 3 **Singh, Vartika**, Prantika Das, Miodrag Stevanovic, et al. "An Inclusive agri-food systems transformation pathway for India." *R and R in Communications Sustainability - Nature Portfolio*, 2026.
- 4 **Singh, Vartika**, Ankit Saha, Ranjan Kumar Ghosh, and Aline Mosnier. "India's Roadmap to net-zero through the AFOLU Sector." *Under Preparation for submission to PLoS One*, 2026.
- 5 **Singh, Vartika**, Ankit Saha, Udit Rana, Sowmiya Subramaniam, and Ranjan Kumar Ghosh. "A sub-national approach to explore sustainable food and land-use systems for India." *Submitted to Agricultural Economics*, 2026.

Policy Briefs and Discussion Papers

- 1 **Singh, Vartika**, Abhijeet Mishra, et al. "Biofuels in India: Trading off climate mitigation with water security goals." 2025.

- 2 Kishore, Avinash, and **Vartika Singh**. "Seeds, water, and markets to increase wheat productivity in Bihar, India." 2021.
- 3 Jha, Chandan Kumar, Ranjan Kumar Ghosh, Satyam Saxena, **Vartika Singh**, Miodrag Stevanović, Alexander Popp, and Hermann Lotze-Campen. "Pathways to Sustainable Land-Use and Food Systems, 2020 Report of the FABLE Consortium." 2020.
- 4 Kishore, Avinash, **Vartika Singh**, and Shweta Gupta. "Sticky Seeds: Why Old Seeds Continue to Dominate the Rice-Wheat Agriculture in Eastern India." 2020.
- 5 Ceballos, Francisco, Samyuktha Kannan, **Vartika Singh**, and Berber Kramer. "Digital Technologies for financial inclusion of smallholder farmers: Needs assessment in three states of India." 2019.
- 6 Ghosh, Ranjan Kumar, Shweta Gupta, **Vartika Singh**, and Patrick S Ward. "Is there a market for multi-peril crop insurance in developing countries moving beyond subsidies? Evidence from India." Discussion Paper Intl Food Policy Res Inst, 2019.
- 7 **Singh, Vartika**, and Sujata Ganguly. "Evaluation of India's Soil Health Card from User's Perspective." 2018.
- 8 **Singh, Vartika**, and Patrick Ward. "Challenges in implementing India's Aadhaar-enabled fertilizer management system." 2018.
- 9 **Singh, Vartika**, Patrick Ward, and Shweta Gupta. "Who are the entrepreneurs? A case for accelerated service economy for agriculture machinery in Odisha." 2018.
- 10 Ward, Patrick, Shweta Gupta, **Vartika Singh**, et al. "What is the true value of fertilizer? An assessment of farmers willingness-to-pay for fertilizers across the hill and Terai regions of Nepal." 2014.
- 11 Ward, Patrick, David Ortega, David Spielman, and **Vartika Singh**. "Farmer preferences for drought tolerance in inbred and hybrid rice: Evidence from Bihar, India." 2014.

Book Chapters and Reports

- 1 Ankit, Saha, Singh Vartika, Prantika Das, Aditi Joshi, Chandan Kumar Jha, and Ranjan Kumar Ghosh. *FABLE Scenathon 2023 Results: Pathways for food and land-use systems in India*. Research Report. Paris, France: SDSN, 2024.
- 2 **Singh, Vartika**, Miodrag Stevanović, Ranjan Kumar Ghosh, Benjamin Leon Bodirsky, Alexander Popp, and Hermann Lotze-Campen. *How to reduce agrifood systems' future hidden costs? A multi-country case study -Background report for the State of Food and Agriculture 2024*. Research Report. Paris, France: SDSN, 2024.
- 3 Vittis, Yiorgos, Alison Mosnier Aline; Smith, Ricardo Arguello, John Chavarro Diaz, **Vartika Singh**, et al. *Synthesis of the agrifood systems' hidden costs analysis in the six FABLE country case studies*. Research Report. Paris, France: SDSN, 2024.

Blog Posts

- 1 Saha, Ankit, **Vartika Singh**, and Ranjan Kumar Ghosh. "Why India's food and agricultural systems hold the key to climate action." FABLE Consortium Blog, January 15, 2026.
<https://hdl.handle.net/10568/180661>.
- 2 **Singh, Vartika**, and Arabinda Padhee. "Reimagining Indian agriculture for a low-carbon future." IFPRI Blog, December 23, 2025. <https://hdl.handle.net/10568/179626>.
- 3 **Singh, Vartika**, and Claudia Ringler. "Groundwater conservation and food security: Navigating pathways to sustainable agriculture." IFPRI Blog, July 15, 2024.
<https://hdl.handle.net/10568/170243>.
- 4 Lautze, Jonathan, Mathew McCartney, and **Vartika Singh**. "The WEF nexus and One Health: Complex links present challenges and opportunities." CGIAR, April 17, 2024.
<https://hdl.handle.net/10568/145087>.

Select International Conferences

- 1 *Food Security in the Indo-Gangetic Basin*. Delivering for Nutrition in South Asia, 2025, Nepal.
- 2 *An inclusive agri-food systems transformation approach for India*. International Conference of Agricultural Economists (ICAE 2024), 2024, New Delhi.
- 3 *An inclusive agri-food systems transformation approach for India*. Delivering for Nutrition in South Asia, 2024, Sri Lanka.
- 4 *Evaluating Regional Diets against Global Recommendations for Sustainable Food System Transformation in India*. Royal Geographic Society -Annual International Conference 2024, 2024, London.
- 5 *A bottom-up approach to sustainable transformation: accounting for regional preferences in global environmental goals*. Tropentag, 2023, Berlin.
- 6 *An inclusive agri-food systems transformation approach for India*. CGIAR Gender Impact Platform Conference on From Research to Impact: Towards just and resilient agri-food systems, 2023, New Delhi.
- 7 *Dietary Transitions As Climate Mitigating Measures and Their Environmental Impacts in South Asia*. Agricultural & Applied Economics Association (AAEA), 2023, Online.
- 8 *The role of food systems in achieving India's sustainability targets: Analysis using integrated assessment modelling for India*. Agriculture, Nutrition, and Health (ANH) Academy, 2022, Online.
- 9 *Water Use and Crop Production in India: Our 2 Cents*. Washington State University, 2022, Seattle.
- 10 *Groundwater Conservation and Global Food Security: An exploratory assessment*. 2022, Sustain Valencia.
- 11 *Seeds, Water, and Markets to Increase Wheat Productivity in Bihar, India*. 31st International Conference of Agricultural Economists (ICAE), 2021, Online.

Invited Presentations

June 2025	2050 Is Now Global Convening funded by IKI, Berlin Invited to speak on India's Low-emission Development Pathways by Food, Agriculture, Biodiversity, Land-Use and Energy (FA-BLE) Consortium Wageningen University and Research, Netherlands Invited to present my work on exploring global and regional scenarios for groundwater conservation and food security
Nov 2023	College of Agriculture and Life Sciences (CALS), Cornell University Invited to present my research on sustainable transformation pathways for India: The case of water, food and land
Mar 2024	MANAGE, Hyderabad Invited to speak on Climate Change and its implications on Agriculture & Food Security at National Institute of Agricultural Extension Management (MANAGE)(An organization of Ministry of Agriculture & Farmers Welfare Govt. of India)

Projects and Grants

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| 2025 | World Bank Collaborated with World Bank Central Asia team for scenario assessment of irrigation expansion in the Central Asian region. |
| 2024 | Belmont Forum - Climate, Environment and Health Grant II Co-wrote this successful proposal NSF grant from Belmont Forum for 100,000 Euro for project called SOLVE - System adaptation for OneHealth under cLimate change for Vulnerable groups and Ecosystems. |

Projects and Grants (continued)

	OECD Collaborated with OECD to develop future climate scenarios on groundwater use for global food baskets. Findings contributed to the flagship report of the Global Commission on the Economics of Water.
2023	FAO Collaborated with FAO to develop India country case study on hidden costs of food systems, as part of their on annual flagship State of Food and Agriculture (SOFA 2024) report. Food Systems Economics Commission Scientific collaboration on this project with teams from PIK, Oxford University, Stockholm Resilience Centre and EAT to develop pathways for economics, social, climate and health scenarios for India.
2019- present	FABLE Consortium Co-led activities for the India country team, in collaboration with organizations from 29 countries to develop the first suite of models to assess India's policy pathways within the agricultural, food and landuse sectors.
2012-2017	BMGF and USAID Co-led activities on Cereal Systems Initiative for South Asia, collaborative project between IFPRI, CIMMYT and IRRI - a decade long project partnership supporting empirical research in collaboration with state governments.

Capacity Development

2014 onwards	Digital Data Collection I regularly conduct several sessions with Ministry of Agriculture and Ministry of Rural Development on digital data collection using SurveyCTO and CAPI.
2015-2017	Capacity building of civil servants Organized and led the training sessions of bureaucrats at the Lal Bahadur Shastri National Academy of Administration (LBSNAA), and the Nepal Administrative Staff College (NASC), equipping civil servants with tools on impact evaluation and evidence-based policymaking.
2017-2019	Ministry of Agriculture and Farmer's Welfare Co-designed evaluation(s) with State Departments of Agriculture of Bihar, Odisha, Andhra Pradesh and Uttar Pradesh, to test and redesign of soil health cards, including machinery subsidies and Direct Benefit Transfer (DBT) schemes.
2017-present	Guest Faculty - Impact Evaluation. Annual Course on Corporate Social Responsibility organized by the Ministry of Corporate Affairs.
2021-present	Guest Faculty - Climate Change and Health. Annual Course on Understanding Public Health Nutrition: Towards Policy and Action organized by CCDS, Emory University and IFPRI.
2012 onwards	Policy Stakeholder Engagement Over the years, I have conducted numerous stakeholder engagements with officials from Ministry of Fertilizers, Ministry of Rural Development, Niti Aayog, Comptroller and Auditor General of India and Ministry of Health on key policy priorities of the states.

Peer Review & Editorial Roles

Journal Referee

- Communications: Earth & Environment
- Soil Use and Management
- Scientific Reports (Nature Portfolio)
- Environmental Research Letters
- Food Policy

Guest Editor

- Springer Nature collection — *Green Technology, Resilience, and Sustainability*

Awards and Associations

2019 – present	Guest Scientist , Potsdam Institute for Climate Impact Research
2019 – 2024	Contractual Collaboration , Indian Institute of Management, Ahmedabad
2023	NSF - Early Career Research Award , Awarded the competitive ECR grant (6000 USD) by Global to Local Analysis of Systems Sustainability (GLASSNET) program to pursue research and collaboration with Cornell University
2021-2023	Member of Advisory Board . Food, Agriculture, Biodiversity, Land-Use and Energy Consortium - across 23 countries
2024-present	Chairperson-Workplace Committee For IFPRI's South Asia Regional Office over 3 offices and more than 100 staff.
2018	Capacity Building Award IFPRI - for undertaking capacity building both within and outside the Institute.